软测第四次

```
d:\Desktop>pytest test_calculator.py
============================================= test session starts =============================================
platform win32 -- Python 3.11.7, pytest-7.4.0, pluggy-1.0.0
rootdir: d:\Desktop
plugins: cov-6.1.1, anyio-4.2.0
collected 4 items

test_calculator.py ....                                                                              [100%]

============================================== 4 passed in 0.04s ==============================================
```

命令提示符

```
D:\anaconda\Lib\site-packages\_pytest\python.py:617: in _importtestmodule
    mod = import_path(self.path, mode=importmode, root=self.config.rootpath)
D:\anaconda\Lib\site-packages\_pytest\pathlib.py:565: in import_path
    importlib.import_module(module_name)
D:\anaconda\Lib\importlib\__init__.py:126: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
<frozen importlib._bootstrap>:1204: in _gcd_import
    ???
<frozen importlib._bootstrap>:1176: in _find_and_load
    ???
<frozen importlib._bootstrap>:1147: in _find_and_load_unlocked
    ???
<frozen importlib._bootstrap>:690: in _load_unlocked
    ???
D:\anaconda\Lib\site-packages\_pytest\assertion\rewrite.py:169: in exec_module
    source_stat, co = _rewrite_test(fn, self.config)
D:\anaconda\Lib\site-packages\_pytest\assertion\rewrite.py:351: in _rewrite_test
    tree = ast.parse(source, filename=strfn)
D:\anaconda\Lib\ast.py:50: in parse
    return compile(source, filename, mode, flags,
E     File "d:\Desktop\test_calculator.py", line 6
E               return a - b
E       ^
E   TabError: inconsistent use of tabs and spaces in indentation
=========================================== short test summary info ===========================================
ERROR test_calculator.py
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
============================================== 1 error in 0.25s ===============================================

d:\Desktop>
```

```
D:\Desktop\新建文件夹>pytest --cov=calculator --cov-report=html
============================================= test session starts =============================================
platform win32 -- Python 3.11.7, pytest-7.4.0, pluggy-1.0.0
rootdir: D:\Desktop\新建文件夹
plugins: cov-6.1.1, anyio-4.2.0
collected 4 items

test_calculator.py ....                                                                              [100%]

================================================ tests coverage ================================================
_____________________________________ coverage: platform win32, python 3.11.7-final-0 _____________________________________

Coverage HTML written to dir htmlcov
============================================== 4 passed in 0.16s ==============================================
```

Coverage report: 100%

Files | Functions | Classes

filter...

hide covered

coverage.py v7.8.0, created at 2025-05-22 21:41 +0800

File ▲	class	statements	missing	excluded	coverage
calculator.py	Calculator	6	0	0	100%
calculator.py	*(no class)*	5	0	0	100%
Total		**11**	**0**	**0**	**100%**

coverage.py v7.8.0, created at 2025-05-22 21:41 +0800

Coverage report: 91%

Files | Functions | Classes

filter...

hide covered

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

File ▲	class	statements	missing	excluded	coverage
calculator.py	Calculator	6	1	0	83%
calculator.py	*(no class)*	5	0	0	100%
Total		**11**	**1**	**0**	**91%**

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

Coverage report: 91%

Files | Functions | Classes

filter...

hide covered

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

File ▲	statements	missing	excluded	coverage
calculator.py	11	1	0	91%
Total	**11**	**1**	**0**	**91%**

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

Coverage report: 91%

Files | Functions | Classes

filter...

hide covered

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

File ▲	function	statements	missing	excluded	coverage
calculator.py	Calculator.add	1	1	0	0%
calculator.py	Calculator.subtract	1	0	0	100%
calculator.py	Calculator.multiply	1	0	0	100%
calculator.py	Calculator.divide	3	0	0	100%
calculator.py	*(no function)*	5	0	0	100%
Total		**11**	**1**	**0**	**91%**

coverage.py v7.8.0, created at 2025-05-22 21:54 +0800

```
D:\Desktop\新建文件夹>pytest test_todo.py
============================================= test session starts =============================================
platform win32 -- Python 3.11.7, pytest-7.4.0, pluggy-1.0.0
rootdir: D:\Desktop\新建文件夹
plugins: cov-6.1.1, anyio-4.2.0
collected 10 items

test_todo.py ..........                                                                              [100%]

============================================== 10 passed in 0.05s =============================================
```

```
D:\Desktop\新建文件夹>pytest --cov=todo_list --cov-report=html
============================================= test session starts =============================================
platform win32 -- Python 3.11.7, pytest-7.4.0, pluggy-1.0.0
rootdir: D:\Desktop\新建文件夹
plugins: cov-6.1.1, anyio-4.2.0
collected 13 items

test_calculator.py ...                                                                               [ 23%]
test_todo.py ..........                                                                              [100%]

================================================ tests coverage ================================================
_____________________________________ coverage: platform win32, python 3.11.7-final-0 _____________________________________

Coverage HTML written to dir htmlcov
============================================== 13 passed in 0.15s =============================================
```

Coverage for **todo_list.py**: 100%

32 statements | 32 run | 0 missing | 0 excluded

« prev ^ index » next coverage.py v7.8.0, created at 2025-05-22 22:52 +0800

```python
class Todo:
    def __init__(self, name, completed=False):
        self.name = name
        self.completed = completed

    def mark_completed(self):
        self.completed = True

    def __str__(self):
        return f"{self.name} - {'Completed' if self.completed else 'Pending'}"

class TodoList:
    def __init__(self):
        self.todos = []

    def add_task(self, name):
        if not name:
            raise ValueError("Task name cannot be empty")
        self.todos.append(Todo(name))

    def remove_task(self, name):
        task_to_remove = None
        for task in self.todos:
            if task.name == name:
                task_to_remove = task
```

```
D:\Desktop\新建文件夹>pytest --cov=test_data_processor --cov-report=html
============================================= test session starts =============================================
platform win32 -- Python 3.11.7, pytest-7.4.0, pluggy-1.0.0
rootdir: D:\Desktop\新建文件夹
plugins: cov-6.1.1, mock-3.14.0, anyio-4.2.0
collected 15 items

test_calculator.py ...                                                                               [ 20%]
test_data_processor.py ..                                                                            [ 33%]
test_todo.py ..........                                                                              [100%]

================================================ tests coverage ================================================
_____________________________________ coverage: platform win32, python 3.11.7-final-0 _____________________________________

Coverage HTML written to dir htmlcov
============================================== 15 passed in 0.43s =============================================
```

Coverage for **test_data_processor.py**: 100%

25 statements | 25 run | 0 missing | 0 excluded


```python
# test_data_processor.py
import pytest
from calculator import Calculator
from data_processor import DataProcessor

@pytest.fixture
def calculator():
    """Fixture: 提供Calculator实例"""
    return Calculator()

@pytest.fixture
def data_processor(calculator):
    """Fixture: 提供DataProcessor实例"""
    return DataProcessor(calculator)

def test_process_data_success(mocker, data_processor):
    """
    测试process_data方法在成功获取数据时的行为
    """
    # Mock requests.get方法
    mock_get = mocker.patch('data_processor.requests.get')
    mock_response = mocker.MagicMock()
    mock_response.status_code = 200
    mock_response.json.return_value = [1, 2, 3, 4]
    mock_get.return_value = mock_response
```